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Optimizing Software Requirements Selection: A Hybrid K-Means and Intuitionistic Fuzzy TOPSIS Approach

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***Abstract*—The success of a software development project relies greatly on the priority given to the software requirements. Such a process is however easy to capture because evaluation of this requirements is extremely prone to personal differences as well as risk. Our work offers a new contribution to the development of the software requirements prioritization process. This approach incorporates K-means clustering and intuitionistic fuzzy AHP. It classifies the similar requirements using the clustering technique, while the vague aspect of human perception is addressed using multi-criterion decision making MCDM. Practical aspects of the method are demonstrated with an illustrative example of requirement prioritization for an Examination System (ES). The results emphasize the potential of the approach in improving the transparency and accuracy of decision-making fluidity, in which the project managers can adopt it as a strategic tool for planning and resource distribution. Nevertheless, practical limitations have been met, and reliance on expert judgement requires therefore further work to eliminate these constraints. All in all, this hybrid approach provides a strong basis for handling the challenges of prioritization of software requirements to provide better processes of software development that is responsive to changes.**

Keywords: Software Requirements Selection; K-Means Clustering; Intuitionistic Fuzzy Set; Fuzzy TOPSIS; MCDM; Functional Requirements; Non-Functional Requirements

I. Introduction

Software development seems to have enjoyed the rapid growth in the development of software due to the changing technologies in software engineering [1]. This has made it difficult for project managers and the developers to make correct decisions when it comes to prioritizing software requirements. This is because priorization is done to guarantee that the maximum amount of useful and useful features is completed within the time, budget, and resource constraints [2, 10]. Yet it is also an exercise that is very difficult because the needs of the clients are diverse and views on the importance of some of the requirements are rather subjective [3].

In this manner, traditional techniques in software requirements prioritization do not appreciate the different dimensions of the various requirements in a condition where prioritization is needed [4]. Such tools most of the time do not accurately capture the fuzzy concept of people's preferences leading to wrong decisions being made [5]. moreover, the most complex factor is also the prioritization of non-functional requirements (NFRs): security, usability, performance due to a lack of quantitative measure of these issues and a high degree of inter relatedness with stakeholders' qualitative evaluation. This work presents a novel hybrid method to combine IFT and K-means clustering to assist in the software requirements prioritization tasks even more efficiently [6]. The K-means clustering method used here is rather innovative in that it reformats numerically expressed data into structured forms that communicate similar ideas, by clustering like demands through matrix and language participations. This clustering step reduces the evaluation activities and therefore the complexity of making decision is facilitated by structural organisation of the activities.

Interior Intuitionistic Fuzzy sets (IFS) intelligence system includes the concept of disaggregation and can be viewed as an extension of IFS theory, a more versatile one [7]. In discerning connections, relationships or when hard data is not present. It is primarily useful in cases where there is a lack of precision for opinions from various stakeholders, that is, the opinions are foggy [8]. The proposed [9] advantages in terms of decision making using It contains an international perspective on management decision-making IFT emphasizes fuzzy sets. This section explains how non-functional requirements (NFRs) are methodically incorporated into functional requirements (FRs) analysis o basis software requirements. This aids decision-makers in requirements prioritization and project objectives realignment.

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The proposed method's practicality and effectiveness are demonstrated through a case study of an ES. The findings demonstrate how fuzzy decision-making and clustering improve stakeholder participation while more successfully highlighting accuracy and transparency. This work gives a significant addition to the field of software requirements prioritization by providing a systematic method for ranking the software requirements using the advantages of the clustering method and the Intuitionistic Fuzzy Set to manage the complexity present in modern software development projects. The following is a list of the research's contributions:

- A unique combination of the Intuitionistic Fuzzy TOPSIS and the K-means clustering is presented in this work that enhance the precision and robustness of the software requirements prioritization.
- An in-depth assessment of software requirements is achieved by considering both Functional requirements (FRs) and NFRs.
- The approach provides a systematic way of addressing ambiguity and uncertainty in requirement descriptions through the application of fuzzy logic and clustering.
- Although the proposed approach is flexible and adaptable to several domains where software requirements prioritizing is important, the study concentrates on an Examination System.

II. Research Methodology for Software Requirement Selection in an Examination System

This study investigates the effectiveness of combining KMeans clustering and IFT to select the most important software requirements for an ES. Initially, all potential requirements were identified. Then, appropriate methods were chosen, and a research approach was designed. Figure 1 illustrates the research methodology used in this study to select software requirements.

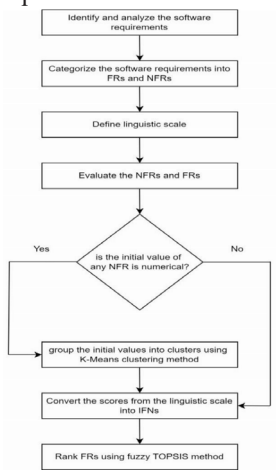


Fig. 1: Research Methodology for Software Requirement Selection in an Examination System

A. Research Methodology

Step 1: Identify and Analyze the Requirements

Identify and analyze the necessary software requirements.

Categorize the software requirements into FRs and NFRs.

Step 2: Define Linguistic Scale

Establish a linguistic scale for rating the data using intuitionistic fuzzy numbers (IFNs).

Step 3: Evaluate the NFRs and FRs.

Assess the NFRs and FRs using a predefined linguistic scale.

Step 4: Group Numerical Data

If the initial value of a NFR is a number, categorize it into groups using the K-Means clustering algorithm.

Step 5: Convert Scores

Convert the linguistic ratings into IFNs.

Step 6: Rank Alternatives

Use the fuzzy TOPSIS method to rank the FRs

III. Discussion and Result

A. Data Requirements

This paper has already outlined the methodology. Here, we provide a sample calculation for selecting software requirements of an ES. The NFRs used are Budget (in USD) (NFR1), Security (NFR2), Usability (NFR3), and Reliability (NFR4).

B. Determination of Linguistic Scale

Tables I and II present the linguistic scale used for evaluating NFRs and FRs. This scale aids in assigning values to both criteria and software requirements through linguistic terms or reasoning. The values on this scale are expressed using IFS. The IFS weights will be applied in the next ranking phase.

Table I: A linguistic scale to evaluate the weights of NFRs.

Scale		IFS Numbers
Extremely Important	EI	(0.90, 0.10)
Important	I	(0.70, 0.25)
Moderately Important	MI	(0.55, 0.40)

Table II: A linguistic scale is used to evaluate FRs in relation to NFRs.

Scale		IFS Numbers
Very High	VH	(0.90, 0.10)
High	H	(0.75, 0.20)
Low	L	(0.40, 0.50)

C. Assigning Weights and Evaluating the Requirements

Table III presents the evaluation of NFRs and FRs using linguistic scales. This evaluation involves the initialization of the linguistic scale.

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D. Clustering Data with K-Means

The valuation data in Table III includes a budget represented as a numerical data type. To convert this numerical data into a single linguistic scale, the K-Means clustering process is applied. The following outlines the steps involved in the clustering process.

Step 2: Initialization of the center of cluster.

The cluster center should be initialized to zero. The centroid, or central point of a cluster, represents the average of all the data points within that cluster. You can assign a distinct starting point to each cluster by using either the maximum or minimum value from the dataset. Equations 1 and 2 can be used to calculate the centroid of each cluster.

$$cC1 = \min(B1, B2, \ldots\ldots, Bn) \quad (1)$$

$$cC2 = \max(B1, B2, \ldots\ldots, Bn) \quad (2)$$

where,

cC1 and cC2 represent the center points of clusters C1 and C2, while B1- Bn are the budget data points that need to be grouped.

$$cC2 = min\ (15,13,15,13,12,12,14,12,15,15) = 12$$

$$cC2 = max\ (15,13,15,13,12,12,14,12,15,15) = 15$$

Step3: Identify the data to the closest cluster

Calculate the distance between each data point and every cluster center using equation 3. The resulting distances are summarized in Table IV.

$$Dis(i,j) = \sqrt{(B1i - B1j)^2 + \cdots + (Bki - Bkj)^2} \quad (3)$$

where,

$Dis(i,j)$ = Distance of data I to centroid j

Bki = Data I on FR data to k

Bkj = Centroid j on FR i

Step 4: Identify the centroid using the clustering results from Step 3

The centroid for each cluster is obtained by calculating the average of the data points within the cluster, as illustrated in equations 4 and 5. The newly computed centroids, based on the distance values from Table IV, are displayed in Table V.

$$cC1n = \frac{sum(B1 \ldots Bn\ C1)}{n} \quad (4)$$

$$cC2n = \frac{sum(B1 \ldots Bn\ C2)}{n} \quad (5)$$

$$cC1n = \frac{sum(13,13,12,12,12)}{5} = \frac{62}{5} = 12.4$$

$$cC2n = \frac{sum(15,15,14,15,15)}{5} = \frac{74}{5} = 14.8$$

Step 5: Reassign all data points to their nearest cluster based on the updated cluster centers

The clustering process continues until the groups of data points (clusters) stop changing. If the clusters remain the same after a round, the process is finished. If not, the process starts again from step 3. Table 5 shows how the budget data was categorized using a linguistic scale. NFR C-1 was rated as High, while NFR C-2 was rated as Low.

Table III: Software Requirements Assessment

	NFR_1	NFR_2	NFR_3	NFR_4
Rating/FRs	EI	I	I	I
FR_1	15	H	L	H
FR_2	13	H	L	H
FR_3	15	H	L	H
FR_4	13	H	VH	VH
FR_5	12	VH	VH	VH
FR_6	12	VH	VH	VH
FR_7	14	VH	L	H
FR_8	12	VH	VH	VH
FR_9	15	VH	L	H
FR_{10}	15	H	L	H

Table IV: Calculating the Spatial Relationship of FRs to the Centroid

	Distance to C1	Distance to C2	Result
FR_1	3	0	C-2
FR_2	1	2	C-1
FR_3	3	0	C-2
FR_4	1	2	C-1
FR_5	0	3	C-1
FR_6	0	3	C-1
FR_7	2	1	C-2
FR_8	0	3	C-1
FR_9	3	0	C-2
FR_{10}	3	0	C-2

Table V: The findings are Categorized According to Linguistic Levels

	Distance to C-1	Distance to C-2	Result	Linguistic Scale
FR_1	2.6	0.2	C-2	L
FR_2	0.6	1.8	C-1	H
FR_3	2.6	0.2	C-2	L
FR_4	0.6	1.8	C-1	H
FR_5	0.4	2.8	C-1	H
FR_6	0.4	2.8	C-1	H
FR_7	1.6	0.8	C-2	L
FR_8	0.4	2.8	C-1	H
FR_9	2.6	0.2	C-2	L
FR_{10}	2.6	0.2	C-2	L

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Table VI: Ratings to IFS numbers conversion

NFRs/ FRs	NFRs			
	NFR1 (0.90, 0.10)	NFR_2 (0.70, 0.25)	NFR3 (0.70, 0.25)	NFR4 (0.70, 0.25)
FR_1	(0.40, 0.50)	(0.75, 0.20)	(0.40, 0.50)	(0.75, 0.20)
FR_2	(0.75, 0.20)	(0.75, 0.20)	(0.40, 0.50)	(0.75, 0.20)
FR_3	(0.40, 0.50)	(0.75, 0.20)	(0.40, 0.50)	(0.75, 0.20)
FR_4	(0.75, 0.20)	(0.75, 0.20)	(0.90, 0.10)	(0.90, 0.10)
FR_5	(0.75, 0.20)	(0.90, 0.10)	(0.90, 0.10)	(0.90, 0.10)
FR_6	(0.75, 0.20)	(0.90, 0.10)	(0.90, 0.10)	(0.90, 0.10)
FR_7	(0.40, 0.50)	(0.90, 0.10)	(0.40, 0.50)	(0.75, 0.20)
FR_8	(0.75, 0.20)	(0.90, 0.10)	(0.90, 0.10)	(0.90, 0.10)
FR_9	(0.40, 0.50)	(0.90, 0.10)	(0.40, 0.50)	(0.75, 0.20)
FR_{10}	(0.40, 0.50)	(0.75, 0.20)	(0.40, 0.50)	(0.75, 0.20)

E. Conversion Process for IFS numbers

The criteria weights and software requirements, which were initially described using linguistic terms, were transformed into IFS numbers. The converted values are presented in Table 1 and Table 2. The results of this conversion for criteria and software requirements are shown in Table VI.

F. Process of Ranking Using Fuzzy TOPSIS Method

After converting the data into IFS numbers, we will use the fuzzy TOPSIS method to identify the most important functional requirements. The process involves these steps:

Step 1: Generate a fuzzy decision matrix

Equation 6 is employed to reduce the IFS number to a solitary value, simplifying the computational process.

$$V(i,j) = (i - \max(i)) * 2 + (j - \min(j)) * 2 + ((1 - (\max(i) + \min(j))) - (1 - ((\max(i) - \min(j)))) * 2 \quad (6)$$

Conversion for NFR1:

$$V(0.90,0.10) = (0.90 - 0.90) * 2 + (0.10 - 0.10) * 2 + ((1 - (0.90 + 0.10)) - (1 - (0.90 - 0.10))) * 2 = 0.40$$

Conversion for NFR2, NFR3, and NFR4:

$$V(0.70,0.25) = (0.70 - 0.90) * 2 + (0.25 - 0.10) * 2 + ((1 - (0.90 + 0.10)) - (1 - (0.90 - 0.10))) * 2 = 0.30$$

Conversion for FR1, FR3, FR7, FR9, and FR10 with respect to NFR1 is illustrated below:

$$V(0.40,0.50) = (0.40 - 0.75) * 2 + (0.50 - 0.20) * 2 + ((1 - (0.75 + 0.20)) - (1 - (0.75 - 0.20))) * 2 = 0.70$$

Conversion for FR2, FR4, FR5, FR6, FR8 with respect to NFR1 is illustrated below:

$$V(0.75,0.20) = (0.75 - 0.75) * 2 + (0.20 - 0.20) * 2 + ((1 - (0.75 + 0.20)) - (1 - (0.75 - 0.20))) * 2 = 0.80$$

Conversion for FR1, FR2, FR3, FR4, FR10 with respect to NFR2 is illustrated below:

$$V(0.75,0.20) = (0.75 - 0.90) * 2 + (0.20 - 0.10) * 2 + ((1 - (0.90 + 0.10)) - (1 - (0.90 - 0.10))) * 2 = 0.30$$

Conversion for FR5, FR6, FR7, FR8, FR9 with respect to NFR2 is illustrated below:

$$V(0.90,0.10) = (0.90 - 0.90) * 2 + (0.10 - 0.10) * 2 + ((1 - (0.90 + 0.10)) - (1 - (0.90 - 0.10))) * 2 = 0.40$$

Conversion for FR1, FR2, FR3, FR7, FR9, FR10 with respect to NFR3 is illustrated below:

$$V(0.40,0.50) = (0.40 - 0.90) * 2 + (0.50 - 0.10) * 2 + ((1 - (0.90 + 0.10)) - (1 - (0.90 - 0.10))) * 2 = 0.20$$

Conversion for FR4, FR5, FR6, FR8 with respect to NFR3 is illustrated below:

$$V(0.90,0.10) = (0.90 - 0.90) * 2 + (0.10 - 0.10) * 2 + ((1 - (0.90 + 0.10)) - (1 - (0.90 - 0.10))) * 2 = 0.40$$

Conversion for FR1, FR2, FR3, FR7, FR10 with respect to NFR4 is illustrated below:

$$V(0.75,0.20) = (0.75 - 0.90) * 2 + (0.20 - 0.10) * 2 + ((1 - (0.90 + 0.10)) - (1 - (0.90 - 0.10))) * 2 = 0.30$$

Conversion for FR4, FR5, FR6, FR8 with respect to NFR4 is illustrated below:

$$V(0.90,0.10) = (0.90 - 0.90) * 2 + (0.10 - 0.10) * 2 + ((1 - (0.90 + 0.10)) - (1 - (0.90 - 0.10))) * 2 = 0.40$$

The inverse value or divergen value (d) of each single value (v) is computed by using the eq. 7. All the inverse values are presented by matrix M.

$$d = 1 - v \quad (7)$$

$$M = \begin{bmatrix} 0.3 & 0.7 & 0.8 & 0.7 \\ 0.2 & 0.7 & 0.8 & 0.7 \\ 0.3 & 0.7 & 0.8 & 0.7 \\ 0.2 & 0.7 & 0.6 & 0.6 \\ 0.2 & 0.6 & 0.6 & 0.6 \\ 0.2 & 0.6 & 0.6 & 0.6 \\ 0.3 & 0.6 & 0.6 & 0.7 \\ 0.2 & 0.6 & 0.6 & 0.6 \\ 0.3 & 0.6 & 0.8 & 0.7 \\ 0.3 & 0.7 & 0.8 & 0.7 \end{bmatrix}$$

Step 2: Create a normalized fuzzy decision matrix

To normalize the fuzzy decision matrix, apply equation 8 to each value in the matrix.

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$$M_{ij} = \frac{x_{ij}}{\sum_{i=1}^{m} x_{ij}^2} \quad (8)$$

where,

M_{ij} =Result of fuzzy normalization of M

$i = 1,2, \ldots, m$; and $j = 1,2, \ldots, n$

$$M = \begin{bmatrix} 0.138462 & 0.115294 & 0.128 & 0.111872 \\ 0.061538 & 0.115294 & 0.128 & 0.111872 \\ 0.138462 & 0.115294 & 0.128 & 0.111872 \\ 0.061538 & 0.115294 & 0.072 & 0.082192 \\ 0.061538 & 0.084706 & 0.072 & 0.082192 \\ 0.061538 & 0.084706 & 0.072 & 0.082192 \\ 0.138462 & 0.084706 & 0.072 & 0.111872 \\ 0.061538 & 0.084706 & 0.072 & 0.082192 \\ 0.138462 & 0.084706 & 0.128 & 0.111872 \\ 0.138462 & 0.115294 & 0.128 & 0.111872 \end{bmatrix}$$

Step 3: Create a weighted normalized fuzzy decision matrix

To determine the weighted importance of each criterion in the fuzzy decision matrix, apply equation 9 to adjust the normalized values.

$$M_{ij} = x_{ij} * w_{ij} \quad (9)$$

$$M = \begin{bmatrix} 0.055385 & 0.034588 & 0.0384 & 0.033562 \\ 0.024615 & 0.034588 & 0.0384 & 0.033562 \\ 0.055385 & 0.034588 & 0.0384 & 0.033562 \\ 0.024615 & 0.034588 & 0.0216 & 0.024658 \\ 0.024615 & 0.025412 & 0.0216 & 0.024658 \\ 0.024615 & 0.025412 & 0.0216 & 0.024658 \\ 0.055385 & 0.025412 & 0.0216 & 0.033562 \\ 0.024615 & 0.025412 & 0.0216 & 0.024658 \\ 0.055385 & 0.025412 & 0.0384 & 0.033562 \\ 0.055385 & 0.034588 & 0.0384 & 0.033562 \end{bmatrix}$$

Step 4: Compute fuzzy positive ideal solution and fuzzy negative ideal solution

Using Equations 10 and 11, as shown in Table VII, we calculated the fuzzy positive ideal solution (PIS) and the fuzzy negative ideal solution (NIS).

$$F^+ = (f_1^+, f_2^+, \ldots, f_n^+) \quad (10)$$

$$F^- = (f_1^-, f_2^-, \ldots, f_n^-) \quad (11)$$

where,

$f^+ = max\ i_{ij}(benefit); min\ i_{ij}\ (cost)$

$f^- = max\ i_{ij}(benefit); min\ i_{ij}\ (cost)$

Table VII: fuzzy PIS and fuzzy NIS

	NFR1	NFR2	NFR3	NFR4
F+	0.055385	0.034588	0.0384	0.033562
F–	0.024615	0.025412	0.0216	0.024658

Step 5: Compute the closeness coefficient

To determine the closeness coefficient, the distance between each fuzzy reference point (FR) and both the fuzzy PIS and fuzzy NIS was calculated using equations 12 and 13, respectively.

$$D_i^+ = \sqrt{\sum_{j=1}^{n} (f_i^+ - f_{ij})^2} \quad (12)$$

$$where\ i = 1,2, \ldots, m$$

$$D_i^- = \sqrt{\sum_{j=1}^{n} (f_{ij} - f_i^-)^2} \quad (13)$$

$$where\ i = 1,2, \ldots, m$$

the preference value of each FR (P_i) is computed using eq. 14.

$$P_i = \frac{D_i^-}{D_i^+ + D_i^-} \quad (14)$$

Step 6: Determine the ranking of FRs

The FRs were prioritized based on their closeness coefficient values, as shown in Table 8.

Table VIII: Ranking order of FRs

FRs	D^+	D^-	CC	Ranking
FR1	0.011648	0.016118	0.58049413	5
FR2	0.010284	0.019471	0.654377416	4
FR3	0.011768	0.016119	0.57801126	6
FR4	0.004183	0.020262	0.828881162	3
FR5	0.002879	0.022009	0.884321761	1
FR6	0.002913	0.020939	0.877871877	2
FR7	0.021732	0.000912	0.04027557	9
FR8	0.015599	0.021025	0.574077108	7
FR9	0.030044	0.0021046	0.065464748	8
FR10	0.023132	0.0000912	0.003927107	10

IV. Conclusion and Future Scope

A. Conclusion

This research proposes a new combined method using KMeans clustering and IFT to rank software requirements. This approach tackles the challenges of evaluating both functional and non-functional requirements by combining the benefits of clustering and multi-criteria decision-making. Our methodology demonstrated how numerical data, such as budgets, can be transformed into linguistic scales through KMeans clustering, thus making them more amenable to humancentric evaluation methods like Intuitionistic Fuzzy TOPSIS. The application of this approach to the selection of software requirements for an ES highlighted several key advantages:

1. By incorporating IFS, our approach effectively captured the uncertainty and vagueness associated with human judgments, allowing for more robust decision-making.
2. By integrating fuzzy decision-making with clustering, the software requirements were prioritized more accurately and effectively, reflecting the actual preferences and priorities of the stakeholders.
3. Even though this study focused on an ES, the proposed approach is adaptable and applicable to several areas where software requirements prioritizing is essential.

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The results indicated that our hybrid approach not only enhances the accuracy of software requirements prioritization but also gives a systematic framework to deal with the complexities of software development projects. The project managers may use the rankings obtained through our method to help with resource allocation, risk management, and strategic planning, ultimately leading to more successful project outcomes.

B. Future Scope

Our work provides a framework for prioritizing the software requirements, even though several research directions need to be explored. First, the clustering phase can be further enhanced by exploring the integration of additional machine-learning methods. Secondly, carrying out empirical research in various industrial environments would confirm the efficiency and applicability of the proposed approach. Lastly, taking realtime feedback from stakeholders during the process of prioritization could dynamically refine requirements based on evolving project requirements and conditions. Overall, the hybrid K-Means and Intuitionistic Fuzzy TOPSIS approach, taken as a whole provides a promising direction for addressing the complex problems of software requirements prioritizing and opens the door for more informed and flexible software development processes.

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